

SARFRAJ SHAIK

sarfrajshaik03@gmail.com | Portfolio Website

PROFESSIONAL SUMMARY

Final-year Computer Science student & aspiring Data Scientist with hands-on experience in machine learning, statistics-driven predictive modeling, robust data engineering, and interactive analytics dashboards.

TECHNICAL SPECIALIZATIONS

Machine Learning & Deep Learning: PyTorch, Scikit-Learn, Predictive Analysis, Computer Vision, CNNs
Data Engineering & Analytics: Python, Pandas, NumPy, SQL, ETL pipelines, Interactive Visualizations
Web & Database Systems: React/Vite, TypeScript, TailwindCSS, Express.js, PostgreSQL, FastAPI

CAPSTONE RESEARCH PROJECTS

1. ClinicaVision: Explainable Radiographic AI Classifier

Designed ensemble neural networks with vision transformers and convolutional layers to identify diagnostic patterns. Implemented layer activation maps (Grad-CAM) to explain local statistical decision weights.

2. AeroSense: Spatiotemporal Environmental Forecaster

Built telemetry pipelines ingesting atmospheric sensor readings. Coded multi-step LSTM & multivariate Prophet modules to forecast micro-ambient air quality changes across urban target zones.

3. DriftSentry: Automated ML Drift Pipeline

Developed modular statistical container assessing pipeline drift and Population Stability Index (PSI) values. Configured auto-retraining alerts triggered upon real-time validation decay.

ACADEMIC & RESEARCH EXPERIENCE

Undergraduate Researcher & Data Science Intern

University CS Research Lab | 2023 - Present

Formulates evaluation metrics, performs exploratory log cleaning, and compiles key ML models.

Computer Science Candidate

School of Info Tech & Engineering | 2020 - 2024

Extensive academic curriculum in advanced algorithms, big data query schemas, and neural network design.

Data Systems Developer

CS Engineering Lab Projects | 2021 - 2023

Engineered modular relational databases, configured web API layers, and created responsive analytics displays.

CORE COMPETENCIES & COMPUTING SKILLS

Development Stack: JavaScript, TypeScript, React, Express, Python, FastAPI, SQL, PostgreSQL, Git, CSS/Tailwind

Data Stack: NumPy, Pandas, Scikit-Learn, PyTorch, Matplotlib, Prophet, ETL Pipelines, Statistical Modeling

Prepared on behalf of Sarfraj Shaik. Fully compliant high-fidelity digital reproduction.